

A STUDY ON THE WELDABILITY OF NiTi SUPERELASTIC ALLOY

A PROJECT REPORT

Submitted in partial fulfillment of the requirements for the degree of

Bachelor of Technology

in

Mechanical Engineering

by

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SCHOOL OF MECHANICAL ENGINEERING



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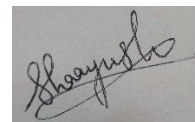
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I hereby declare that the project report entitled “**A study on the weldability of NiTi superelastic alloy**” submitted by me to Vellore Institute of Technology University, Vellore in partial fulfillment of the requirement for the award of the degree of **Bachelor of Technology in Mechanical Engineering** is a record of bonafide project work carried out by me under the supervision of **Prof. Anoj Giri**. I declare that this report represents my concept written in my own words and where other’s ideas or words have been included. I have adequately cited and referenced the original sources. I further declare that I have adhered to all principles of academic honesty and integrity and have not misinterpreted or fabricated or falsified any idea/data/fact/source in my submission. I understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed. Further I affirm that the contents of this report have not been submitted and will not be submitted either in part or in full, for the award of any other degree or diploma and the same is certified.

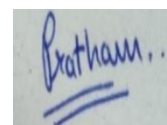
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BONAFIDE CERTIFICATE

This is to certify that the project report entitled “**A STUDY ON THE WELDABILITY OF NiTi SUPERELASTIC ALLOY**” submitted by **SHASHWAT AAYUSH (19BME0361), PRATHAM SHRIVASTAVA (19BME0367)** to Vellore Institute of Technology University, Vellore, in partial fulfillment of the requirement for the award of the degree of **Bachelor of Technology in Mechanical Engineering** is a record of bonafide work carried out by him/her under my guidance. The project fulfills the requirement as per the regulations of this institute and in my opinion meets the necessary standards for submission. The contents of this report have not been submitted and will not be submitted either in part or in full, for the award of any degree or diploma and the same is certified.

Anoj Girdi

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Head of the Department

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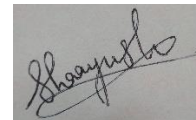
The members of this group would like to thank our professor, Prof. Anoj Giri, for making us explore our innovative sides and come up with such an interesting project. Without your guidance and support our idea would have never been a reality. Thank you, for encouraging us to explore a new field and gain knowledge.

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TABLE OF CONTENTS

CHAPTER NO.	TITLE	PAGE NO.
	EXECUTIVE SUMMARY	i
	LIST OF TABLES	ii
	LIST OF FIGURES	iii
	LIST OF SYMBOLS AND ABBREVIATIONS	iv
1	INTRODUCTION AND LITERATURE REVIEW	
	1.1 Introduction	1
	1.1.1 Motivation	1
	1.1.2 Background	1
	2.1 Literature Review	3
	2.2 Gaps in Literature	5
	2.3 Problem Definition	5
	2.4 Objectives	6
2	METHODOLOGY AND EXPERIMENTAL WORK	
	2.1 Methodology	7
	2.2 Experimental Details	8
	2.2.1 TIG Welding	8
	2.2.2 Laser Welding	9
	2.2.3 Microstructure, SEM and EDS	10
	2.2.4 Microhardness	11
	2.2.5 Tensile Test	12
	2.2.6 XRD Analysis	12
3	RESULTS AND DISCUSSION	
	3.1 Microstructure	13
	3.2 Microhardness	17
	3.3 Tensile Test	18
	3.4 XRD Analysis	21

	3.5 Fractography	22
4	CONCLUSION	
	4.1 Contribution to the Literature	23
	4.2 Scope for the future	24
	REFERENCES	25
	PLAGIARISM REPORT	27
	CONTRIBUTION STATEMENT	31
	PROGRAMME OUTCOME DATA SHEET	32

EXECUTIVE SUMMARY

Nitinol is a widely used shape memory alloy. Its functional properties of shape-memory effect and superelasticity along with corrosion and fatigue resistance and biocompatibility helps nitinol to find its application in wide range of industries like aerospace, biomedical, cosmetic, and structural engineering. However, the poor machinability of nitinol limits its application in full-fledged manner. In this work, 2 mm thick nitinol plates were welded using two widely used welding processes viz. Gas tungsten Arc welding (GTAW) and Laser Welding to study its weldability by identifying the metallurgical and mechanical behaviours of the welded samples. Role of input parameters viz. welding current, welding voltage, laser power and travel speed was identified against the output parameters such as depth of penetration, formation of the grains at various zones, microhardness variation and precipitate formation. TIG welding produced wider fusion and heat affected zone as compared to laser welding. Insufficient heat input produced incomplete penetrated welds for two out of three parameters in TIG welding. Laser welding produced fully penetrated welds, however formation of pores was observed in the fusion zone. The area of weld zone was found to decrease as heat input was reduced. There was a clear reduction in the tensile strength of welded sample as compared to the base metal. Highest tensile strength was observed for laser welded sample at parameter of 2kW, 7 mm/sec. Pattern of decreasing hardness was observed from base metal to the fusion zone. However, detection of precipitates through XRD analysis attributed to sudden peaks at fusion zone in hardness profile. Through fractography, dual mode of fracture was observed in the welded sample.

This study summarizes that laser welding produces welds with narrower bead area, which in turn produces lesser defects in the joint. Heat input and shield gas blow distance are one of the two important parameters to be considered to get a defect free weld.

Keywords: *Shape Memory Effect, Superelasticity, Nitinol, Base metal, Heat Affected Zone, Fusion zone, TIG welding, Laser welding*

LIST OF TABLES

Table No.	Title	Page No.
Table 1	Chemical Composition of the Base Metal	8
Table 2	Welding Parameters for TIG Welding	8
Table 3	Welding Parameters for Laser Welding	10
Table 4	Macrographs of TIG Welded Samples	13
Table 5	Macrographs of Laser Welded Samples	16
Table 6	Tensile Test values obtained for various samples	20

LIST OF FIGURES

Figure No.	Title	Page No.
Figure 1	(a) Welding Automation set-up (b) TIG Welded plates	9
Figure 2	(a) Laser Welding Set-up (b) Laser Welded plates	10
Figure 3	(a) Welded sample and its cut-out (b) Laser welded mounted sample (c) TIG Welded mounted sample	11
Figure 4	(a) Dimensions of the Tensile test specimen (b) Base metal tensile test specimen (c) TIG welded tensile test specimen (d) Laser welded tensile test specimen	12
Figure 5	Microstructure at 10x of (a) Base Metal (b) Heat Affected Zone (c) Fusion Zone For optimized parameter of (75A, 17cm/min) TIG Welding	14
Figure 6	(a) Fusion zone-HAZ interface of (60 A, 17cm/min) (b) Fusion zone-HAZ interface of (75 A, 17cm/min) (c) EDS of (60A, 17cm/min) (d) EDS of (75A, 17cm/min)	15
Figure 7	Fusion zone – HAZ – Base Metal interface for laser welded (a) (2kW, 5 mm/sec) (b) (2kW, 7 mm/sec) (c) (2kW, 10 mm/sec)	17
Figure 8	Microhardness profile	17
Figure 9	Stress vs Strain curves	19
Figure 10	XRD plots	21
Figure 11	SEM analysis of fractured surface	22

LIST OF SYMBOLS AND ABBREVIATIONS

MEMS	Micro Electromechanical Systems
SME	Shape Memory Effect
SE	Superelasticity
SMA	Shape Memory Alloys
TIG	Tungsten Inert Gas
GTAW	Gas Tungsten Arc Welding
EDM	Electrical Discharge Machining
EDS	Energy-Dispersive X-ray Spectroscopy
ASTM	American Society for Testing and Materials
SEM	Scanning Electron Microscope
XRD	X Ray Diffraction
HAZ	Heat Affected Zone

CHAPTER 1

INTRODUCTION AND LITERATURE REVIEW

1.1 Introduction

1.1.1 Motivation

One of the highest performance materials that has drawn interest is Shape memory alloy (SMA). Shape memory effect (SME) and pseudoelasticity are the two characteristics of the shape memory alloy. The deformation caused by any force gets removed when heated up to certain temperature is SME and the ability to regain its original shape and size when the load is removed is known as pseudoelasticity. The superelastic characteristics, shape memory effect, biocompatibility, and corrosion resistance of nickel-titanium (NiTi) SMA are well recognized and because of these properties they are widely used in biomedical, aerospace, automotive and fashion industries. However, there are some constraints that restrict the widespread application of this material in many industries. The inadequate machinability and absence of joining Nitinol to other materials and to itself are the major obstacles to manufacture any component or part using this material. The formation of brittle intermetallic compounds makes the joining of Nitinol SMA a technical difficulty as these compounds result in significant loss of weld joint strength since during solidification cracks associated with the dendritic microstructure of the weld metals are created.

1.1.2 Background

Nitinol got its name in 1960, when it was first observed by Buehler and Wiley in Naval Ordnance Laboratory. Since its discovery, researchers have shown deeper interest to use this material in various applications. Nitinol material is a superior choice in the applications such as biomedical, actuators, aerospace, automotive, and MEMS devices due to its unique properties of Super Elastic (SE) and Shape Memory Effect (SME). These functional properties of nitinol come from solid-state diffusionless transition of austenite and martensite phases. Austenite is a stable phase at low temperature whereas martensite is a high temperature stable phase. Recoverable high strain levels can be achieved with nitinol due to stress-induced martensitic transformation. SME provides the ability of returning to parent shape on application of heat to nitinol. Nitinol has a maximum

performance as the practical material when compared with other materials due to its outstanding properties of fatigue strength, corrosion resistance and biocompatibility.

However, full-fledged utilization of nitinol faces certain limitations as a result of its high sensitivity towards heat. As a result, machining and joining of nitinol becomes a very challenging task. This limits nitinol's application into complex shaped tools and other uses. Several joining techniques have been reported by the researchers, namely brazing, resistance welding, TIG welding and Laser welding. For joining micro parts brazing is popular choice. And for macro size joints TIG and Laser welding are suitable options. TIG welding is an easy to operate and economical process of welding that can be used in any form of coupling. Also, it provides stable arc during welding. Laser welding provides high precision and lesser HAZ which makes it a suitable choice for nitinol, which has limited weldability.

In laser Beam welding a laser is used as a concentrated heat source to melt the material which then fuse together when they cool. The laser beam is coherent light with a single wavelength that has only one phase (monochromatic). The laser beam will produce heat when it collides with a surface because of its low beam divergence and high energy content. It is a flexible method since it can perform narrow and deep welds for thicker materials while welding thin materials at high welding speeds.

Still in both TIG and laser welding, fragile intermetallic compounds like Ti_2Ni and Ni_3Ti are formed which results in significant loss of durability from the welded joints. These precipitates accumulate at the grain boundaries and forms a suitable path for crack propagation. Mechanical properties and superior properties of the welded joints are adversely affected due to it. At higher temperatures nitinol reacts with atmospheric oxygen which can cause problems in welding process. Therefore, to avoid the molten pool from reacting with atmospheric gases, it must be adequately protected during welding and cooled until the temperature of 500 degree or lower are obtained.

In this study, TIG welding (Tungsten inert gas welding) and Laser Welding was performed on 2 mm thick nitinol plates. The mechanical and metallurgical characterization was done for both types of welding processes.

2.1 Literature Review

Fusion welding technique has been a popular metal joining method and it is forecasted to continue to be a preferred welding option. Since this method offers highly efficient joints and economic advantages, it is a preferred welding choice across many industries. [1]

Researchers have been trying to implement fusion welding on smart materials. The advent of smart materials has gained interest of researchers towards the fusion welding of these smart materials. The near equi-atomic Ni-Ti alloy, also known by the name nitinol is a popular SMA. The remarkable properties of Shape memory effect and superelasticity makes this a novel material. Due to these properties, nitinol finds its applications in various revolutionary and futuristic purposes.

The biocompatibility of nitinol serves to make stents for damaged vessels and various components of the surgical devices. [3] NiTi suffers a decrease in its mechanical properties after undergoing welding through fusion welding techniques. [2] The ratio between power and welding speed relates to the heat input during welding. The net heat input is a crucial parameter for microstructure.[4]

Deepan Bharathi. et al [4] conducted an experiment to study the effect of laser welding on Nitinol. Seven out of nine experimental runs gave a full penetration. They reported that the temperature gradient resulted into finer grains at the centre of the weld zone with an average grain size of 2.4 to 4.5 μm as compared to other regions of the weldment. As we moved away from the centre of weld zone towards the base metal the cooling rate pattern decreases which resulted into reduction in hardness values. Due to sudden change in temperature, the grains at the centre undergoes grain refinement which results into higher hardness there. The presence of intermetallic phases of Ti_2Ni enhanced the hardness. The combination of higher power with lower welding speed result into values with lower hardness.

Ali Shojaei Zoeram et al [5] studied the effect of welding speed on the laser welded Nitinol. An increase in tensile strength of 104% was achieved as welding speed increased from 1.25 to 1.75mm/s. It was reported that as heat input decreased from 39.4 to 28.1 J/mm due to decrease in welding speed, the size of columnar grain increased from 65.5 to 128 μm signifying that at lower heat input columnar solidification prevails. Also, with increase in welding speed, lesser intermetallic precipitates were formed leading to higher

tensile strengths. The formation of brittle Ti_2Ni phases at the grain boundaries acts as a suitable path for crack initiation and progression.

To explore the usage of Yb: YAG laser source, Sathiya P et.al [6] performed experiment to examine bead geometry, hardness and corrosion resistance of 1 mm thick welded nitinol joints. They found out that bead geometry (bead width, depth of penetration) of the weld mainly depends on the quantum of heat input, rate of heat input, power intensity, rate of heat loss and thermal capacity of the material. They found that with increase in cooling rate from weld zone interface towards the centreline, across the fusion zone solidification mode transformed from planar to cellular and from cellular to dendritic respectively. Cooling rate of the weld is inversely proportional to the heat input. Finer sized grains were obtained at the weld centre as compared to the other regions with an average grain size of 1.1 μm to 3.2 μm due to higher cooling rate at the weld centre. The grain size has a negligible effect due to focal position and shielding gas blown distance. Finer grains gave a higher hardness at the centre of weld zone. The presence of intermetallic phases Ni_3Ti , TiFe_2 , and TiO_2 , gave rise to higher hardness values and hardness decreases due to the presence of coarser grains and pores. They concluded that Yb: YAG laser is a good option for welding Nitinol with improved weld properties.

Kannan T [8] used GRA technique to predict the optimized parameters. The authors reported that in Yb: YAG bead on plate welding, nitinol sheets with higher hardness, depth of penetration and lower bead width are indication of better performance. The average grain size in the weld region was obtained to be 1.6 μm . The authors founded that the average hardness in the optimized NiTi weld zone was 350.1 $\text{HV}_{0.5}$ which is higher than base metal hardness value of 280 $\text{HV}_{0.5}$ due to finer grains in the weld zone. They concluded that welding speed had the greatest impact on the multiple performance characteristic, followed by shielding gas blown distance, power and focal position.

In their other work, Kannan T [7] founded that due to heterogeneities in the weld created the critical issue of change of phase transformation temperature. Formation of micro pores, crack due to poor selection of welding parameters in laser welded samples leads to affect the properties immensely. Weld contained dendritic grains with fine grains at the centre and coarser grains near the interface. The authors confirmed through entropy measurement method that hardness should be given highest weightage as output parameter followed by bead width.

However, there are some constraints that restricts the widespread application of this material in many industries. The inadequate machinability and absence of joining Nitinol to other materials and to itself are the major obstacles to manufacture any component or part using this material. [9]

Fusion welding techniques like electron beam welding, laser beam welding and Tungsten Inert Gas Welding (TIG) are used. Since TIG has numerous benefits such as it has a simpler welding procedure, it is applicable to any form of coupling, simpler to achieve stability of arc even with low current and welding can be done on thin plates. Hence TIG is a crucial welding process.[10]

According to some literature and authors due to formation of large Heat affected zone (HAZ), the GTAW welding technique adversely impacts the mechanical characteristics of welded NiTi Joints. The functional qualities of the welded joints were also restricted by the presence of impurities [11]

However due to formation of brittle intermetallic phases like Ni_3Ti and Ti_2Ni which are formed during the welding significantly lower the weld quality [12]

2.2 Gaps in Literature

Through literature review it was founded that, majority of the analysis of input and output parameter for laser and TIG welding was done for sample size of thickness less than or equal to 1 mm. Very few works have been carried out for higher thickness values of nitinol plates. Higher thickness nitinol plates have structural applications, which can be subjected more loads as compared to lower thickness joints. Also, a comparative analysis between two widely used nitinol joining processes i.e., TIG and laser welding is missing in literature.

2.3 Problem Definition

Both TIG and Laser welding are popular joining techniques for nitinol. However, the formation of brittle intermetallic during in both these processes causes the reduction of mechanical and superior properties of nitinol. Mechanical and metallurgical characterization will be carried for both the types of welding.

2.4 Objectives

- Collect data and information from current findings to identify the right parameters for the welding.
- To evaluate tensile strength, hardness values and microstructures of the welded samples and compare the results for both laser and TIG welding processes.
- To identify the formation of different precipitates at the weld zone.
- To draw out conclusions on how weld zone properties like weld width, weld depth, weld neck width, weld bottom width varies according to different parameters and how it affects the properties of the joints.
- To identify and relate what factors leads to formation cracks, pores, intermetallic, in the weld joint and how they can be minimized.
- To examine the size of different grains formed at different zones, and width of different zones. Investigating the relation of input parameters like solidification rate, heat input, travel speed and power with output parameters like grain size, microhardness, tensile strength, and intermetallic formations.

CHAPTER 2

METHODOLOGY AND EXPERIMENTAL WORK

2.1 Methodology

2.1.1 Literature Study

Data collection from previous works was done. Various welding input parameters Such as welding voltage, welding current, welding speed, inert gases used, composition of the inert gas, flow rate of the inert gas and input power were identified. Suitable tests were selected to test the mechanical and microstructure behaviour of the welded sample. Tensile Test, Vickers Hardness test, optical microscopy, SEM analysis and XRD analysis were decided as required tests.

2.1.2 Material Preparation

A 200mm x 200mm nitinol plate was arranged, which was cut into multiple 25mmx100mm plates to perform welding as per different parameters. Wire EDM process was used to cut the samples.

2.1.3 TIG welding and tests for the corresponding samples

After finalizing the input parameters, firstly TIG welding was performed. After welding the samples were prepared for the corresponding tests, after which individual tests were performed.

2.1.4 Laser Welding and tests for the corresponding samples

After finalizing the input parameters, Laser welding was performed. After welding, the samples were prepared for the corresponding tests, after which individual tests were performed.

2.1.5 Result analysis

After completion of tests for both types of welded samples. Relationship between input and output parameters was analysed and conclusions were drawn.

2.2 Experimental Details

2.2.1 TIG Welding

The material selected for our study was Nitinol super-elastic sheet having thickness 2 mm. The chemical composition for material is shown in Table 1. The samples were cut using the wire EDM process from the initial stock size 200 mm x 200 mm in to multiple 25 mm x 100 mm plates. After cutting, the samples were cleaned using acetone to remove the contaminants from the surface. Proper alignment between the joining surfaces was ensured while fixing the setup for welding. Linear movement automation with pipe fixture setup Figure 1(a) was used to perform the TIG welding of the samples. The weld samples were made with using the filler material and without keeping the root gap. The argon shielding gas flow rate was kept constant i.e., 14 lts /min for all samples. Input variables for the welding were decided based on the past works by the researchers. Table 2 shows the welding parameters used for welding.

Table 1. Chemical composition of the Base Metal

Ni[wt-%]	Ti[wt-%]	Mn[wt-%]	Pd[wt-%]
49.9	50.0	0.019	0.024

Table 2. Welding parameters for TIG welding

Sample	Current (A)	Voltage(V)	Welding Speed(cm/min)
1	60	12	17
2	75	12	17

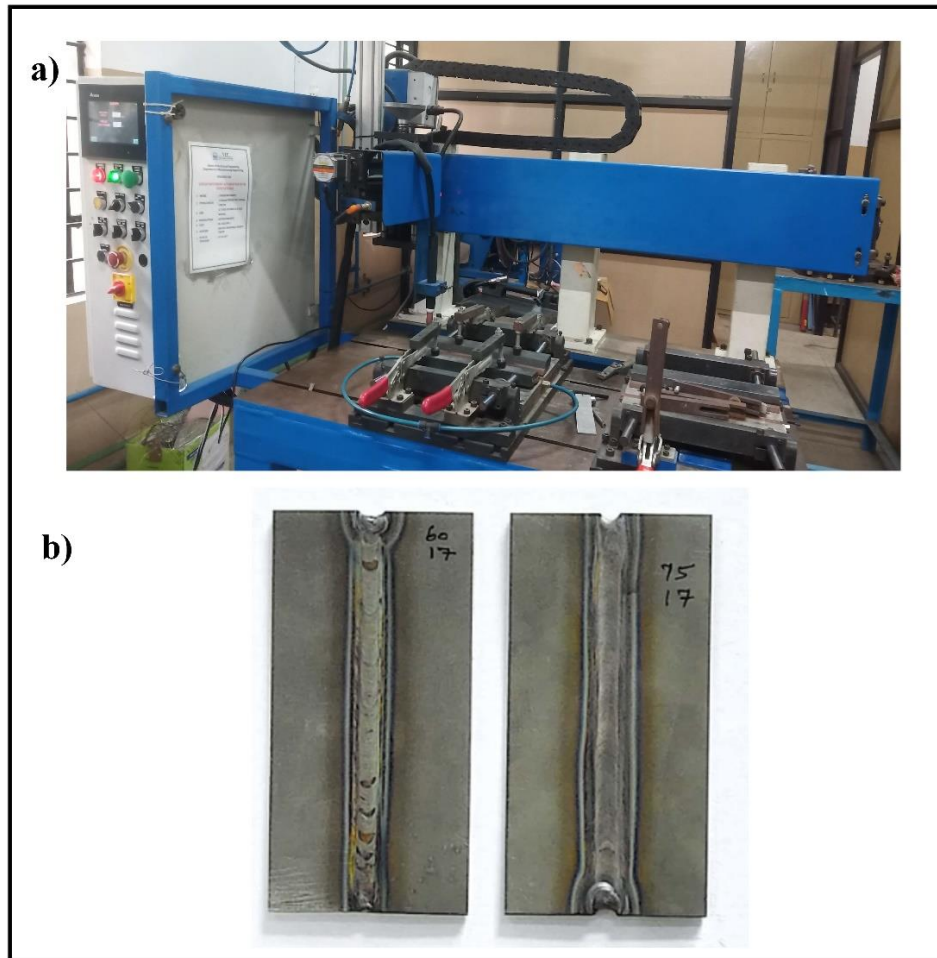


Figure 1: (a) Welding automation set-up (b) TIG Welded plates

2.2.2 Laser Welding

Laser source trumpf 2kW fiber laser welding setup with focal position 100mm, fiber wavelength 1086 nm and fiber beam spot diameter 100 μm was used to perform laser welding of the samples in continuous mode Figure 2(a). Plate dimensions and material composition were kept similar to the ones used in TIG welding. After properly cleaning the samples with acetone and aligning the joining surfaces, welding was performed on the samples with constant laser power and varying scan speed. To prevent the contamination of the molten metal pool by atmosphere, blanket of nitrogen shielding gas with flow rate of 18 lpm was used. Table 3 shows the parameters used for laser welding of the samples. Laser welded plates are shown in figure 2(b).

Table 3. Welding parameters for Laser welding

Sample	Welding Power (kW)	Welding Speed (mm/sec)
1	2	5
2	2	7
3	2	10

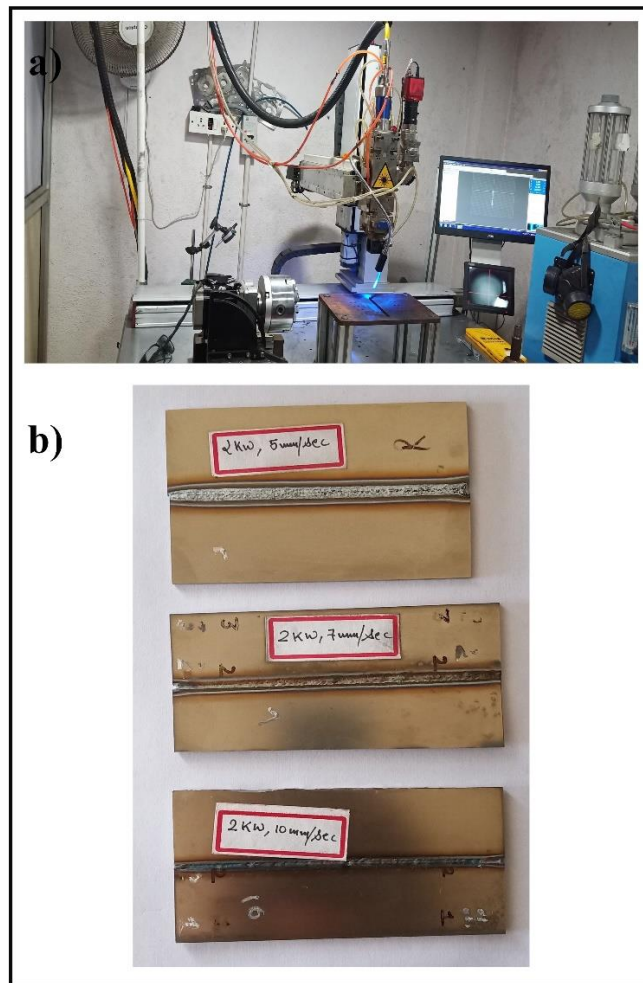


Figure 2: (a) Laser welding set-up (b) Laser Welded plates

2.2.3 Microstructure, SEM and EDS

After the completion of the weld, a small section was cut from the centre of the weldment to examine the after-effects of the welding. Wire EDM process was used to cut the sample to avoid the formation of irregular surface. Figure 3(a) shows the weldment and the cut-out portion used for examination. For the microstructural analysis of the cross-section of

the weld, hot mounting of the cut-out sample was done, exposing the penetrated surface. Figure 3(b) and 3(c) shows the image of the mounted samples for laser and TIG welded samples respectively. The samples were polished using emery sheets of grit size 600, 800, 1200, 1500 and 2000. Cloth polishing using alumina solution followed by diamond polishing gave a mirror polish finish to the metallographic samples. After polishing, samples were etched using HF:HNO₃:H₂O solution with a dilution of 1:5:10 for 30 seconds to expose the grains and zones formed during the welding process. Micrographs were taken for these samples through both optical microscope and SEM technique. EDS analysis was also obtained for the imaged sections.

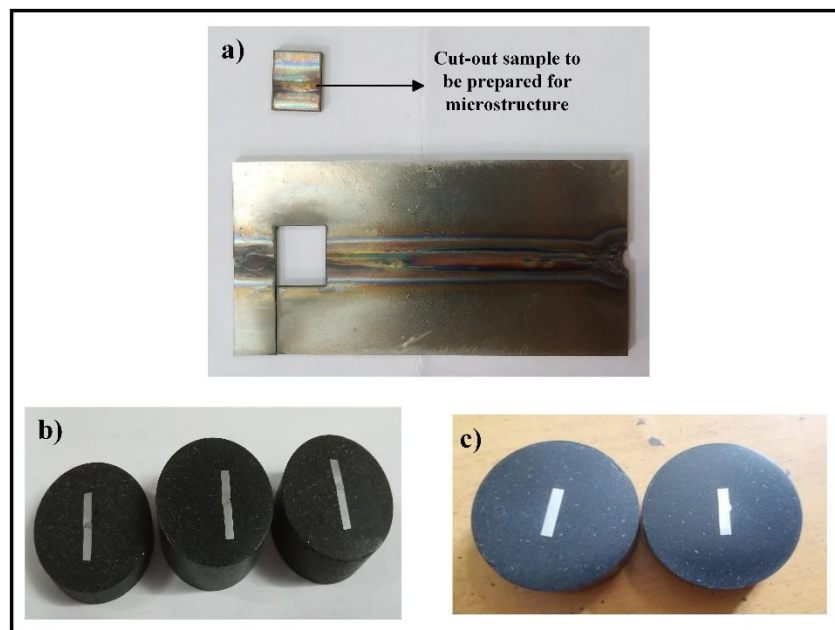


Figure 3: (a) Welded sample and its cut-out (b) Laser welded mounted samples (c) TIG welded mounted samples

2.2.4 Microhardness

To estimate the variation of microhardness along the weld zone, Vicker's microhardness test was conducted on the polished and etched cross sectional surface using Matsuzawa MMT-X Digital Microhardness tester. ASTM E384 standard was used to conduct microhardness test. Load of 50gf was applied for a dwell period of 10 seconds from one end to the other with gap between the adjacent points as 0.25 mm.

2.2.5 Tensile Test

Tensile test was conducted for both base metal and welded sample to analyse the variation of tensile strength after welding. The specimens were designed as per ASTM E8 standard for the tensile test. Outline of the sample with its measurements is shown in figure 4 (a). These samples from the plates were prepared using Wire Electrical Discharge Machining process. Tensile testing of the samples was performed using Instron 8801 Universal Testing Machine. During the test, the samples were loaded with a head speed rate of 0.5mm/min, until they got ruptured. After the completion of the test, the broken samples were collected to analyse the mode of fracture by studying the images of ruptured surfaces using Scanning Electron Microscope (SEM) analysis.

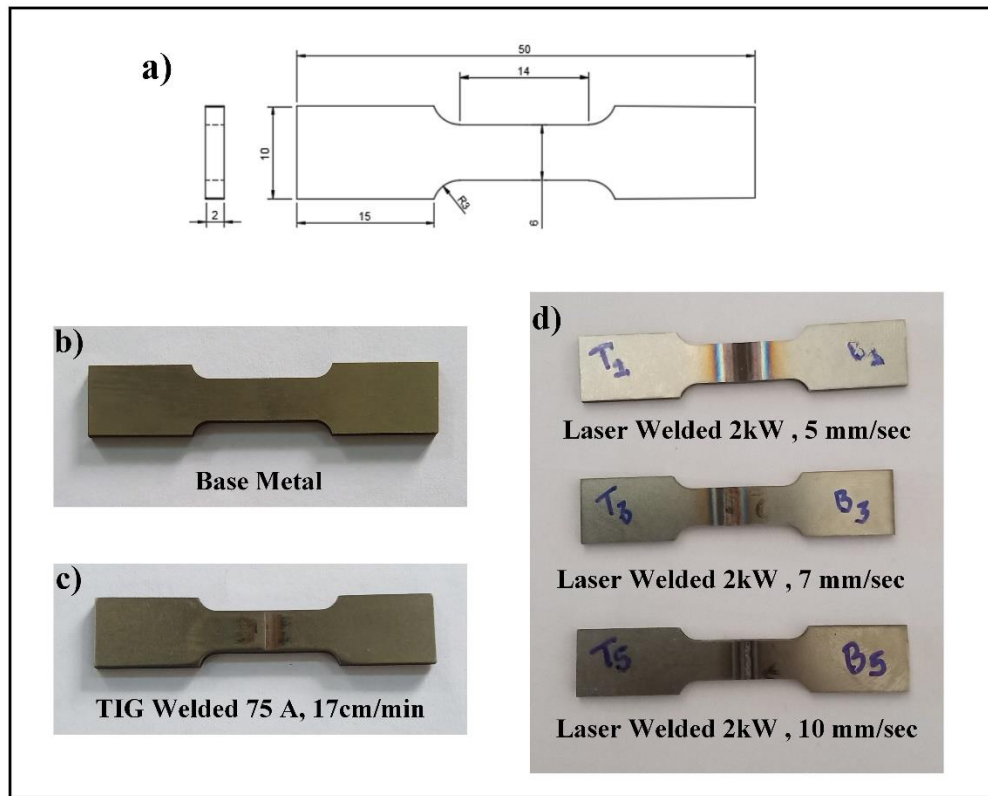


Figure 4: (a) Dimensions of the tensile test specimen (b) Base Metal tensile test specimen (c) TIG welded tensile test specimen at 75 A (d) Laser welded sample tensile test specimen

2.2.6 XRD Analysis

To know about the phases present in the base metal and the precipitates formed after welding in the weld zone, XRD analysis was conducted using powder x-ray diffractometer.

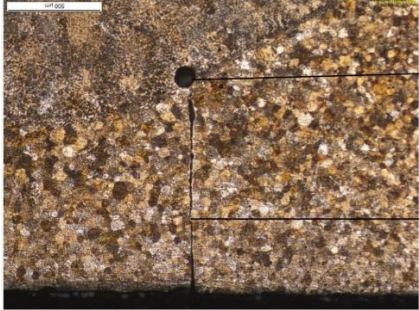
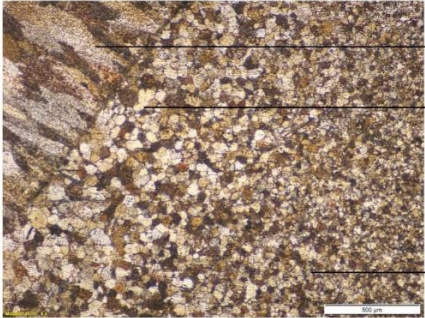
CHAPTER 3

RESULTS AND DISCUSSIONS

3.1 Microstructures

Optical macrographs of the samples at 5x magnification for various parameters of TIG welding are shown in table 4.

Table 4: Macrographs of TIG Welded samples

Sample	Voltage (V)	Current (A)	Travel Speed (cm/min)	Macrographs (5X Magnification)
1	12	60	17	 This macrograph shows a weld cross-section with several dark, circular pores. A vertical line indicates a region of incomplete penetration at the root of the weld. Labels 'Pores' and 'Incomplete Penetration' point to these features.
2	12	75	17	 This macrograph shows a weld cross-section with a clear fusion zone, a heat-affected zone (HAZ), and the base metal. Labels 'Fusion Zone', 'HAZ', and 'Base Metal' point to these regions. A scale bar of 500 µm is visible at the bottom right.
3	12	60	15	 This macrograph shows a weld cross-section with several dark, circular pores. A vertical line indicates a region of incomplete penetration at the root of the weld. Labels 'Pores' and 'Incomplete Penetration' point to these features. A scale bar of 500 µm is visible at the bottom right.

The crack in the macrograph of sample 1 (60A,17 cm/min) indicates incomplete penetration. To check if, the full penetration weld can be achieved, a second trial was conducted by keeping the welding current constant at 60 A and lowering the travel speed to 15cm/min. However, macrograph of sample 3 (60A,15cm/min) still showed some portion of incomplete penetration. Nonetheless, the unwelded portion was seen to reduce as the welding speed was decreased. It is attributed to the fact of increase in heat input (ratio between power and welding speed) as travel speed was reduced. This led to increase in the region of fusion zone.

Complete penetration was observed in Sample 2 (75A, 17 cm/min) along with the formation of distinctive interfaces between the three zones (i.e., fusion zone, Heat Affected Zone (HAZ) and base metal).

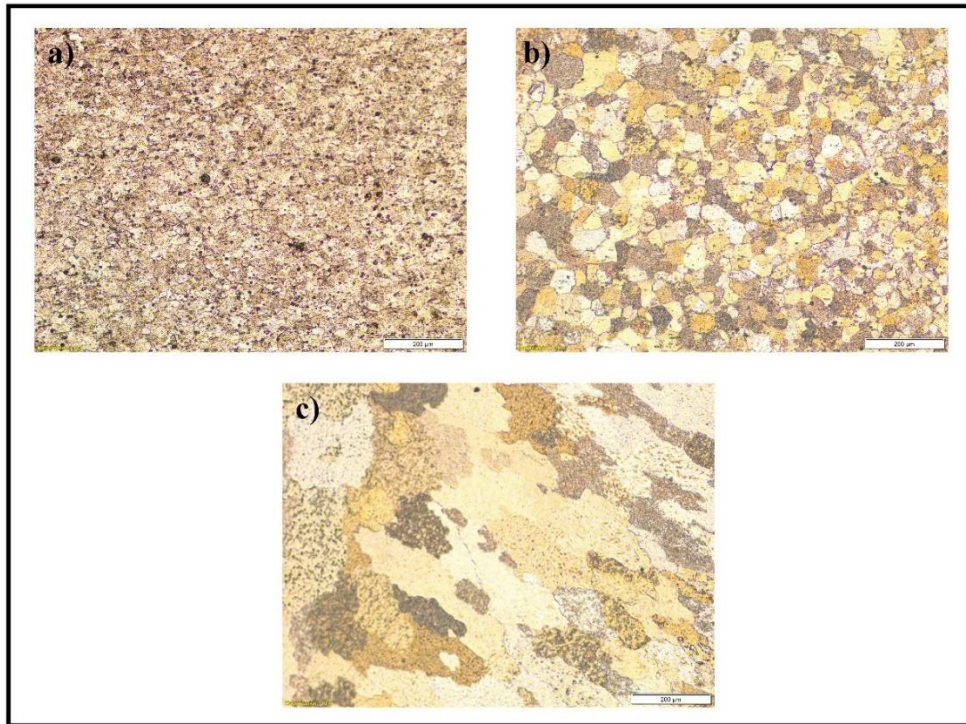


Figure 5: Microstructure at 10x of (a) Base Metal (b) Heat Affected Zone (HAZ) (c) Fusion zone for optimized parameter of (75A, 17cm/min) TIG Welding

Figure 5 shows the image of the three zones of welded joint. Fine sized grains are seen in the micrograph of the base metal figure 5(a) Coarse grains appeared for the Heat Affected Zone Figure 5(b) and elongated grains were seen in the fusion zone figure 5(c). This variation in grain size is mainly decided by the temperature gradient, which is in turn dependent on the supplied heat input. At the centre of the weld bead, heat input is

maximum and cooling rate is minimum which gives more time for grains to solidify. This phenomenon promotes the formation of columnar grains in the fusion zone. At the interface, cooling rate is maximum and thus solidification rate is also maximum which leads to the development of equiaxed grains.

On comparing micrographs from figure 6, sample 1 (60A, 17cm/min) and sample 2 (75A, 17 cm/min), it is observed that with increase in welding current, size of the columnar grains increases. And on comparing micrographs of sample 1(60A, 17cm/min) and sample 3 (60A, 15cm/min), it is observed that with constant welding current, lowering of travel speed promotes enlarged grains. Both of these observations alligns with formation of finer grains with lesser heat input.

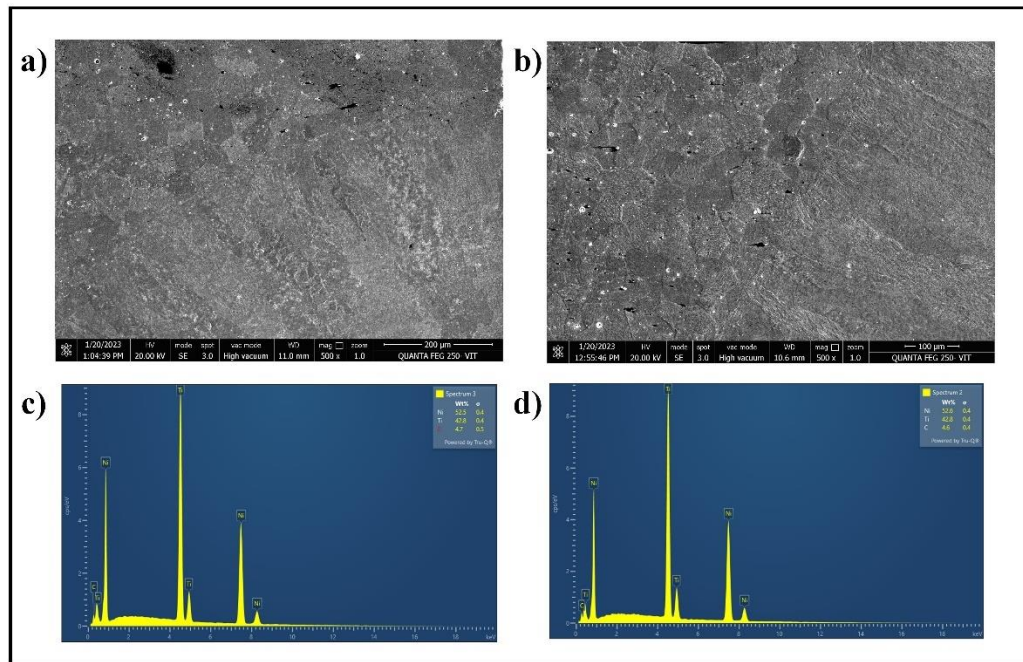


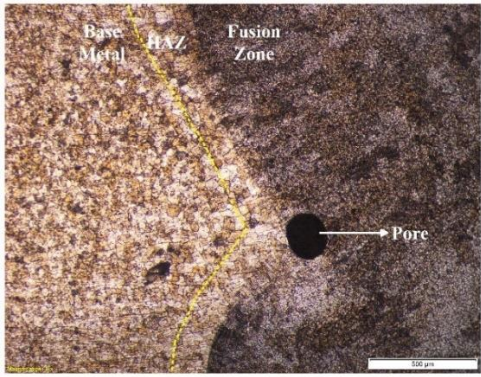
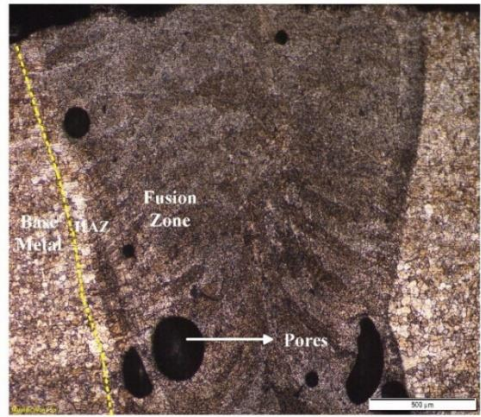
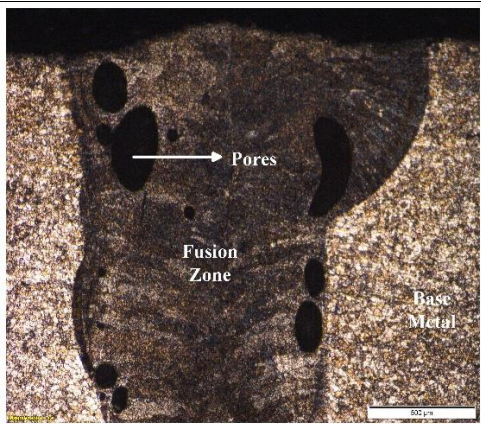
Figure 6: (a) Fusion zone -HAZ interface of (60A, 17 cm/min) (b) Fusion zone -HAZ interface of (75A, 17 cm/min) (c) EDS of (60A, 17 cm/min) (d) EDS of (75A, 17 cm/min)

Table 5 shows the optical macrographs at 5x magnification of the bead profiles obtained for various parameters of Laser welding. In sample 1 (2 kW, 5 mm/sec) larger bead is formed as a result of higher heat input. Distinctive HAZ with maximum width is observed in sample 1. With increment in travel speed, bead profile was found to reduce in size.

With reduction in heat input, the broad top bowl-shaped bead profile is found to change into a narrower bead. The neck region started to appear in sample 2 (2 kW, 7 mm/sec).

As mentioned for TIG welding, heat input is maximum and cooling rate is minimum at centre of the bead which promotes columnar grains. However, on moving towards interface solidification rate increases. Due to this phenomenon, solidification process starts from adjoining layer of partly melted base metal. As these front proceeds towards centre, dendritic structures are formed.

Table 5: Macrographs of Laser Welded samples

Sample	Power(kW)	Travel Speed (mm/sec)	Macrographs (5X Magnification)
1	2	5	 Macrograph of Sample 1 at 2 kW and 5 mm/sec. The image shows a cross-section of the weld with labels for 'Base Metal', 'HAZ' (Heat Affected Zone), and 'Fusion Zone'. A single 'Pore' is indicated by an arrow. A 500 µm scale bar is present in the bottom right corner.
2	2	7	 Macrograph of Sample 2 at 2 kW and 7 mm/sec. The image shows a cross-section of the weld with labels for 'Base Metal', 'HAZ', and 'Fusion Zone'. Multiple 'Pores' are indicated by an arrow. A 500 µm scale bar is present in the bottom right corner.
3	2	10	 Macrograph of Sample 3 at 2 kW and 10 mm/sec. The image shows a cross-section of the weld with labels for 'Fusion Zone' and 'Base Metal'. Multiple 'Pores' are indicated by an arrow. A 500 µm scale bar is present in the bottom right corner.

As shown in the macrographs in table 5, pores have appeared in the weld bead. These pores have appeared as a result of shielding gas blown distance. It could have been avoided if the distance was kept more. Nitrogen gas got trapped inside the molten pool leading to the formation of such pores. However, in sample 1 (2 kW, 5 mm/sec) lesser number of pores were observed as compared to sample 2 (2 kW, 7 mm/sec). It is because the ratio of penetration depth and width of the bead was lower in case of sample 1. The number of pores in sample 3 (2 kW, 10 mm/sec) was maximum as minimum bead width was achieved for this parameter. [6]

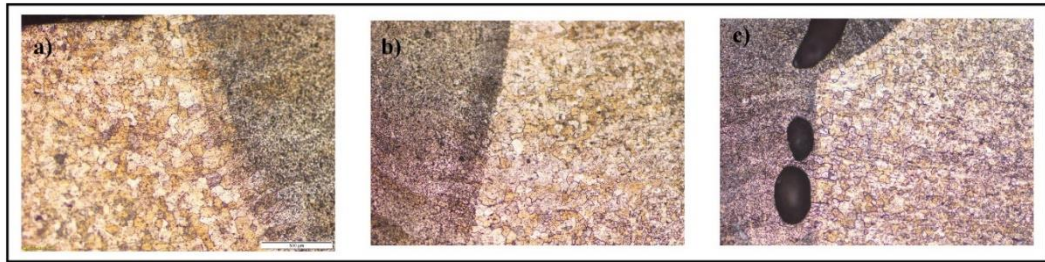


Figure 7: Fusion zone – HAZ – Base metal interface for (a) Sample 1 (2 kW, 5 mm/sec)
(b) (2 kW, 7 mm/sec)

The thickness of the HAZ was found to be greater in sample 1 (2 kW, 5 mm/sec) as compared to sample 2 (2 kW, 7 mm/sec). In sample 3 (2 kW, 10 mm/sec) distinctive interfaces between zones were not visible.

3.2 Microhardness

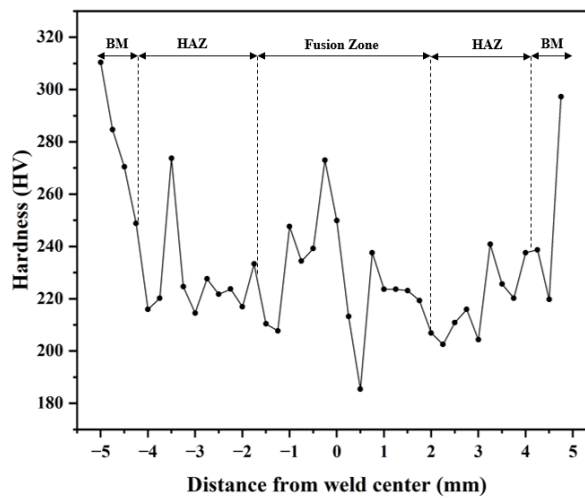


Figure 8: Microhardness profile along the transverse direction of TIG welded sample

Figure 8 shows the microhardness profile across the cross section of the welded sample. In spite of some irregularities, a trend of decreasing hardness from base metal region to fusion zone is observed. Reason for this variation is attributed to grain size at various zones. Fine grains of the base metal led to higher hardness values whereas elongated grains of the fusion zone resulted into lower hardness. Coarser grains forming in the heat affected zone after recrystallization, had intermediate hardness values. Maximum value of microhardness is observed in the base metal because of its fine-grained structure and chemical homogeneity. Due to the appearance of precipitates in the fusion zone, certain rise in hardness values is observed.

3.3 Tensile Test

Stress vs strain curves is shown in figure 9 for base metal and welded samples. The obtained values are tabulated in table 6. Typical stress vs strain curve is obtained for base metal, which shows the superelasticity effect. As load starts to act on the sample, parent phase austenite started to undergo elastic loading. On reaching the maximum load, austenite started to convert into martensite which is shown by plateau in figure 9(a) at around 8% strain. After whole austenite got converted into martensite phase, sudden rise in graph is noticed, indicating elastic loading of martensite phase. On further loading, once again a plateau is observed following which finally rupture occurs.

The tensile test results of the TIG welded sample (75 A, 17cm/min) are shown in figure 9(b). It can be clearly observed from the graph that there is a sudden reduction in tensile strength of the welded sample compared to base metal. The table 6 depicts that maximum strength is of the base material. It can resist tensile stress before fracture up to 1192.46 MPa and the maximum percentage of elongation up to 26.21%. Whereas for the welded sample the maximum strength observed was 548.61 MPa and the elongation at the fracture of 2.783%. Higher percentage of elongation at the fracture for base metal shows the ductile nature of the material. The reduction in the tensile strength of the welded samples is because of the intermetallic phases that gets formed during welding. Also, the coarsening of the grains contributes to the reduction of weld strength.

The tensile test results of the laser welded samples is shown in figure 9(c). It is observed from the graph that as the heat input is increased, there is a slight decrease in the mechanical strength of the welded sample as shown in the table 6. The sample which has

the highest heat input (2kW, 5mm/sec) compared to samples welded at different travel speed has the lowest tensile strain around 3.97 % at the failure. Broader fusion zone formed with these parameters contributed to lesser weld strength. Whereas for sample with welding parameter of (2kW, 7mm/sec) had narrow bead. For sample with welding parameter of (2kW, 10 mm/sec), even narrower bead was achieved. Thus, these parameters produced welds with higher strength than the parameters of (2kW, 5mm/sec). Following the trend, sample with welding parameter of (2kW, 10 mm/sec) should have produced the strongest weld since it had the narrowest weld bead but since maximum number of pores were formed in the bead resulted into lesser strength than sample with welding parameter of (2kW, 7mm/sec).

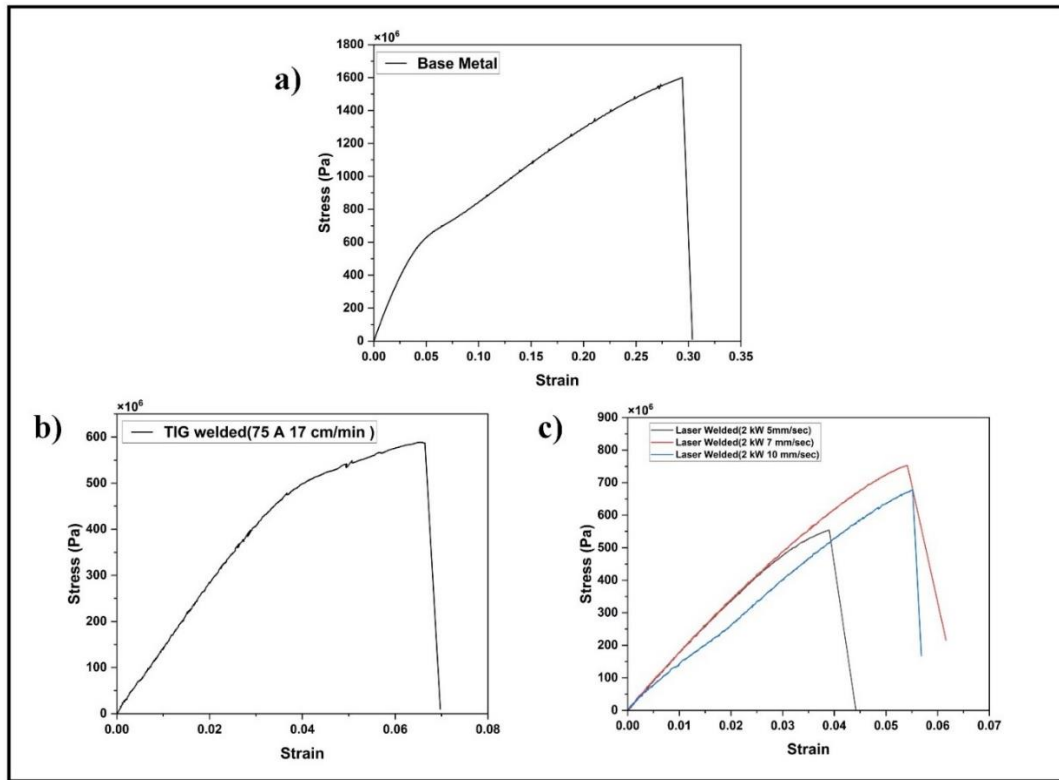


Figure 9: Stress vs strain curve of (a) Base Metal (b) TIG Welded sample (75A, 17cm/min) (c) Laser Welded samples

Both types of welded samples did not show the phase transformation from austenite to martensite as it was showed by the base metal. All of them failed before reaching the mark. The reason for reduction in weld strength is due to multiple factors. Out of which formation of inter metallics and grain coarsening are primary causes. Larger is the heat input, broader will be the formation of weld zone (fusion zone + HAZ). Also, larger heat

input causes more precipitate and pore formation, which hinders the weld strength. As laser welded sample with parameter of (2kW, 5mm/sec), had highest heat input it produced least weld strength. Followed by TIG welded sample (75A, 17 cm/min). Pore formation in the sample of (2kW, 10mm/sec) caused its weld strength to be lesser than the parameter of (2kW, 7mm/sec), otherwise it would have the highest weld strength, since it has a least bead area.

Table 6: Tensile test values obtained for various samples

Sample	Maximum Load (N)	UTS(GPa)	Modulus (Young's) (MPa)	Tensile Stress at Break (MPa)	Tensile Strain at Break (%)	Percentage elongation at Tensile Break (%)
Base Metal	15426.445	1.193	16443.953	1192.46	34.193	26.261
TIG (75A, 17 cm/min)	7137.370	0.552	14014.080	548.61	6.869	2.783
Laser Welded (2kW, 5 mm/sec)	6638.324	0.533	17936.420	532.92	3.977	0.574
Laser Welded (2kW, 7 mm/sec)	9223.604	0.713	17417.158	713.17	5.562	0.91948
Laser Welded (2kW, 10 mm/sec)	8312.332	0.643	13138.285	642.71	5.671	0.38942

3.4 XRD Analysis

Predominantly austenite phase was detected in the parent material as shown in figure 10(a). After welding the formation of intermetallic was observed in the welded sample. Since Nickel was present in majority in base material Ni_3Ti was majorly detected with trace amount of Ti_2Ni .

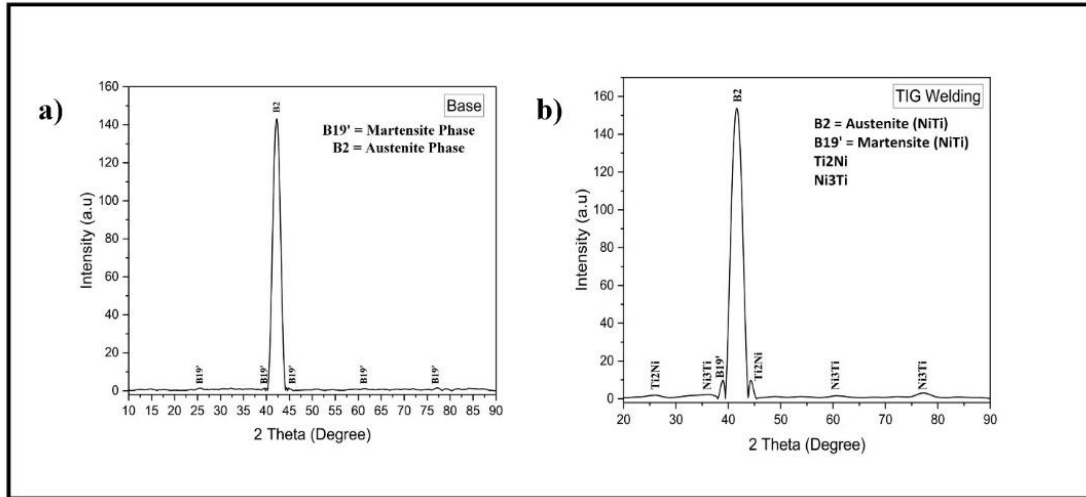


Figure 10: XRD graphs of base metal and TIG Welded sample (75 A, 17cm/min)

3.5 Fractography

Scanning Electron Microscopy (SEM) of the fractured surface of welded samples is shown in the figure 14. Welded sample 14 (c) and 14 (d) there is presence of both dimpled surface and trans granular cleavage which indicates the combination of two modes of failure. Dimple formation indicates the ductile mode of failure whereas intergranular zone indicates brittle mode of failure.

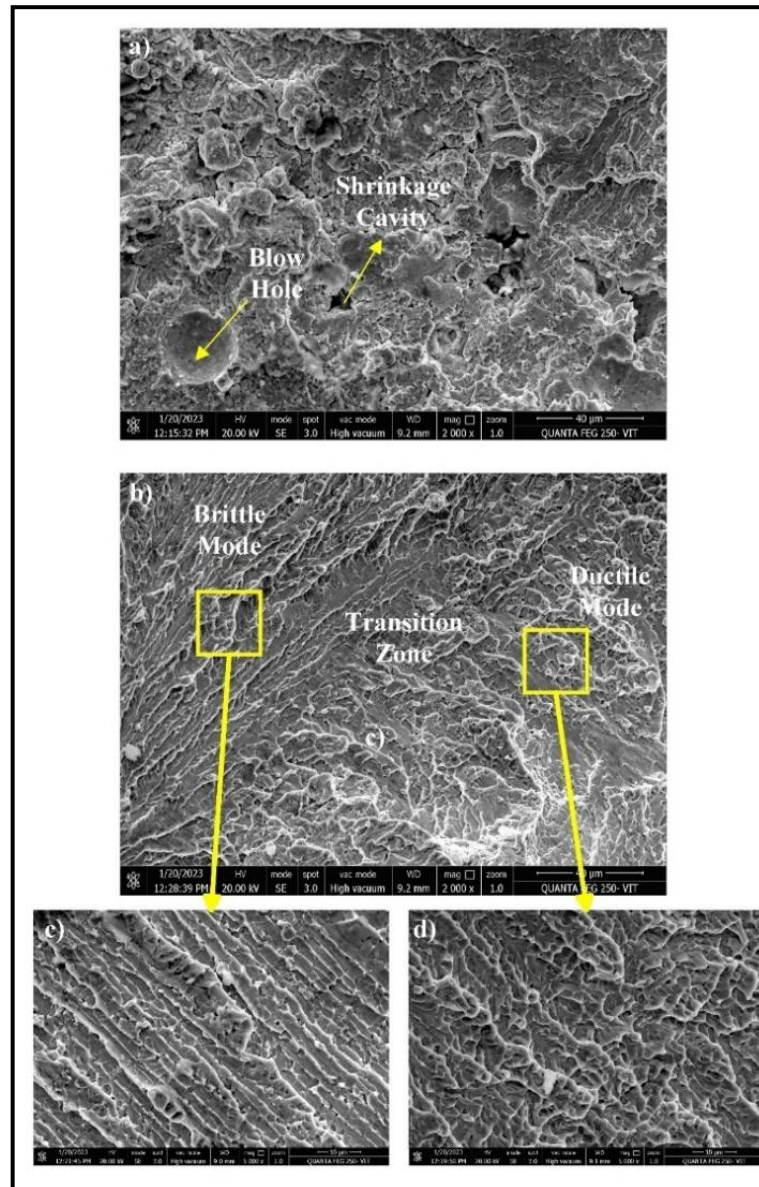


Figure 11. SEM analysis of fractured surface

CHAPTER 4

CONCLUSION

4.1 Contribution to the Literature

TIG welding and Laser welding was performed on 2 mm thick nitinol plates. Following conclusions can be drawn from this study.

- The optical microscopy was performed at three parameters. The parameter of (75A, 17cm/min) produced fully penetrated weld with all three zones clearly visible. The parameter of (60A, 17 cm/min) and (60A, 15 cm/min) did not produce full penetration. It was observed that larger heat input produces elongated grains in the fusion zone.
- Microhardness test was performed on the TIG welded sample and it was observed that hardness was maximum in the base metal and minimum at the fusion zone. It was because of change in grain size from fusion zone to base metal. Abrupt rise in hardness in fusion zone is due to development of precipitates.
- Tensile test of base metal showed the typical stress vs strain graph of nitinol, austenite to martensite phase transfer at the around 8% strain was observed. Weld strength was found least for Laser welded parameter of (2kW, 5mm/sec), followed by TIG (75A, 17 cm/min) , Laser welded parameter of (2kW, 10mm/sec) and Laser welded parameter of (2kW, 10 mm/sec). It was observed that heat input decides the bead shape and profile. Larger heat input leads to larger bead area, which in turn produces welds with lower weld strength.
- Fractography of the TIG welded (75A, 17 cm/min) revealed dimpled surface as well as trans granular cleavage which indicated a combine mode of failure ductile mode and brittle mode respectively.
- Through XRD analysis austenite phase was primarily detected in the base metal. In the welded sample Ni_3Ti was majorly detected with trace amount of Ti_2Ni .
- It can be concluded that laser welding produces welds with narrower weld zone, which leads to formation of lesser precipitates and pores. Heat input and shield gas blow distance are crucial factors for obtaining proper welds. As in this study, insufficient heat input at (60A, 17 cm/min) and (60A, 1 cm/min) was unable to

melt the base metal. Also, there should not be overexpansion of fusion zone due to excess heat input. Shield gas blow distance should be kept in proper range, or else pore formation can take place.

4.2 Scope for the Future

- Conducting laser welding with same parameters and increasing the shield gas blow distance to achieve a pore-free weld.
- Residual stresses analysis in the welded samples.
- DSC analysis of the welded samples.
- Corrosion behaviour of the welded sample.
- Effect of heat treatment on the weldability of nitinol.

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PLAGIARISM REPORT

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CONTRIBUTION STATEMENT

S.No.	Team Member	Registration No.	Contribution
1	Shashwat Aayush	19BME0361	Conducting Literature Review, TIG Welding, Laser Welding, Sample preparation for tensile test and microstructure study using wire EDM, Polishing, Microhardness Tensile Test, Documentation
2	Pratham Shrivastava	19BME0367	Conducting Literature Review, TIG Welding, Sample preparation for tensile test and microstructure study using wire EDM, Polishing, Microhardness Tensile Test, Documentation

Verified by

Anoj Giri

(Dr. Anoj Giri)

PROGRAMME OUTCOMES

Sustainable Development Goals related to the Project

SDG No	SDG Description	Yes/No
1	Eliminate Poverty	No
2	Erase Hunger	No
3	Establish Good Health and Well-Being	No
4	Provide Quality Education	No
5	Enforce Gender Equality	No
6	Improve Clean Water and Sanitation	No
7	Grow Affordable and Clean Energy	No
8	Create Decent Work and Economic Growth	No
9	Increase Industry, Innovation, and Infrastructure	No
10	Reduce Inequality	No
11	Mobilize Sustainable Cities and Communities	No
12	Influence Responsible Consumption and Production	No
13	Organize Climate Action	No
14	Develop Life Below Water	No
15	Advance Life On Land	No
16	Guarantee Peace, Justice, and Strong Institutions	No
17	Build Partnerships for the Goals	No

Justification

Not applicable, the project theme does not match with the given SDG.

Technical / Manufacturing Readiness Level

Select the Domain / Sector (Choose one (or) more appropriate option from the dropdown)

SI No	Domain Description	Yes/No
1	Agriculture & Rural Development innovations	No
2	Consumer Goods and Retail/Supply Chain/ Logistics	No
3	Defence & Security	No
4	Food Processing / Biotech	No
5	Healthcare & Biomedical devices	No
6	Cyber-physical systems, Blockchain, AI & ML Applications	No
7	Industry 4.0	No
8	IoT based technologies	No
9	Manufacturing (Additive Manufacturing, 3D printing, etc.)	No
10	Hydrogen and other alternative fuel technologies.	No
11	New / Innovative Metals and Materials	Yes
12	Other Emerging Areas Innovation for Start-up	No
13	Product Design, Development and Management	No
14	Renewable and Affordable Energy	No
15	Robotics and Drones	No
16	Smart Education	No
17	Smart Vehicles/ Electric vehicle/ Electric vehicle motor and battery technology	No
18	Sustainable Environment	No
19	Waste Management/Waste to Wealth Creation	No
20	Circular economy and zero waste technologies	No

Justification

Smart materials and their fabrication aspects.

Select the Innovation Type

SI No	Innovation Type	Yes/No
1	Product	No
2	Process	Yes
3	Service	No
4	Market Place	No
5	Business/Management	No

Justification

The project deals with the fabrication aspects.

Select the Technical Readiness Level (Choose one (or) more appropriate option from the dropdown)

SI No	Readiness Level	Yes/No
1	Basic research, Principles postulated observed but no experimental proof available	No
2	Technology formulation. The concept and application have been formulated	No
3	Applied research. First laboratory tests completed; proof of concept	Yes
4	Small-scale prototype built in a laboratory environment (ugly prototype)	No
5	Large-scale prototype tested in the intended environment	No
6	Prototype system tested in an intended environment close to expected performance	No
7	Demonstration system operating in the operational environment at a pre-commercial scale	No
8	First-of-a-kind commercial system. Manufacturing issues solved	No
9	Full commercial application, the technology available for consumers	No

Justification

Applied research was done with experimental proof.

Select the Manufacturing Readiness Level (Choose one (or) more appropriate option from the dropdown)

SI No	Readiness Level	Yes/No
1	Basic manufacturing implications identified	No
2	Manufacturing concepts identified	No
3	Manufacturing proof of concept developed	Yes
4	Capability to produce the technology in a laboratory environment	No
5	Capability to produce prototype components in a production-relevant environment	No
6	Capability to produce a prototype system or subsystem in a production-relevant environment	No
7	Capability to produce systems, subsystems, or components in a production-representative environment	No
8	Pilot line capability demonstrated. Ready to begin low-rate production.	No
9	Low rate production demonstrated. Capability in place to begin Full Rate Production.	No
10	Full rate production demonstrated and lean production practices in place	No

Justification

The manufacturing proof was identified.

Course Outcome

Rating of Course Outcomes

CO No	Course Outcome	Rating
1	Formulate specific problem statements for ill-defined real-life problems with reasonable assumptions and constraints	4
2	Perform a literature search and/or patent search in the area of interest	4
3	Develop a suitable solution methodology for the problem	3
4	Conduct experiments / Design & Analysis/solution iterations and document the results	4
5	Perform error analysis/benchmarking / costing	3
6	Synthesise the results and arrive at scientific conclusions/products / solution	3
7	Document the results in the form of technical report/presentation	3

Verified by

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